CSCE 775: Deep Reinforcement Learning and Search

Efficient Point Prompt Optimization for Segmentation using Reinforcement Learning

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1. Introduction

We want to create a reinforcement learning (RL) agent that selects the best point prompts for a segmentation model to yield the best segmentation results for areas of interest in an image. The RL agent should develop an optimal policy to select the best point prompts that act as a target for segmenting select areas in an image. Segmentation models require an abundance of labeled data to achieve good performance. The labeled data requires human annotators which is time consuming. Furthermore, segmentation models still perform poorly on out-of-distribution data samples despite training on large data samples. Few-shot training with a few image samples to generate pseudo labels still requires human edits to correct segmentation masks which is still time consuming.

Prompt-based segmentation models like the Segment Anything Model (SAM) [4] have shown strong zero-shot segmentation performance, but their effectiveness depends heavily on the quality of the input prompts. Optimizing these prompts through RL presents an interesting opportunity to automate and improve the prompt selection process without requiring additional labeled training data.

2. Approach

We plan to develop a Deep Q-Network (DQN) to extend the work by Liu et al. [6]. Liu et al. uses a Q-learning solution to optimize point prompt selection for improved segmentation performance. The environment is a dual-space heterogeneous graph to solve the point prompt optimization problem. The graph is structured as $G = (V, E, \phi, \psi)$ where V is the set of nodes, E is the set of edges, $\phi(v) : V \rightarrow A$ where $A = \{\text{positive nodes } (V_p), \text{ negative nodes } (V_n)\}$, and $\psi(e) : E \rightarrow R$ where $R = \{\text{feature edges } (E_f), \text{ physical edges } (E_p)\}$. The agent uses a physical and feature distance matrix to optimize the point positions based on the geometric relationship between the point prompt and image edge map information. The goal is to control the spread of the positive and negative points while keeping a distance between the positive and negative points. The actions the agent takes to reach the goals is to restore, delete, or move a point. They quantify the states of the graph environment as follows: mean feature distance between positive points in the feature space $D_f^s(p, p)$, mean feature distance between positive and negative points in the feature space $D_f^s(p, n)$, mean feature distance between positive points in the physical space $D_p^s(p, p)$, mean feature distance between negative points in the physical space $D_p^s(n,n)$, and mean feature distance between positive and negative points in the physical space $D_p^s(p,n)$. The reward is penalized when there is an increase in $D_f^s(p,p)$ and $D_p^s(p,n)$. The agent is rewarded if $D_f^s(p,n)$, $D_p^s(p,p)$, and $D_p^s(n,n)$ increase.

We plan to improve upon Liu et al.'s work by implementing a Deep Q-Network (DQN) for a more generalizable policy, training a Graph Neural Network (GNN) to be the policy network encoder for the dual space graph, and incorporating an Intersection over Union (IoU) term in the reward function. A DQN can handle high-dimensional state spaces much better compared to a Q-table solution. Liu et al.'s work heavily compresses the graph whereas a graph-level embedding as a state could help the model learn much more information. Furthermore, the RL agent learns segmentation implicitly through a geometric relationship of the image and feature space without any segmentation-based reward used in the training process. We believe that this could be a major limitation in this solution.

Other related work in this space includes AlignSAM [2], which aligns SAM to open context via reinforcement learning, Matcher [7] for one-shot feature matching, VRP-SAM [9] using visual reference prompts, and PerSAM [11] for one-shot personalization of SAM. Our approach differs from these methods by directly optimizing the point prompt positions through graph-based RL rather than modifying the segmentation model itself.

3. Evaluation

We plan to evaluate our model's segmentation performance and the relationship between segmentation performance and the number of data samples between RL algorithms and supervised learning algorithms. Our baselines for comparison will be between previous RL and supervised segmentation models [2, 6, 7, 9, 10, 11]. We will use the mean Dice Similarity Coefficient (mDSC) and mean Hausdorff Distance (mHD) to evaluate the segmentation model performance between the previous RL and supervised segmentation models. Next, we will conduct a learning curve analysis where we evaluate the models' mDSC and mHD when trained on different amounts of data (25%, 50%, 75%, 100%).

We plan to benchmark our model on two natural and two medical image datasets. The natural image datasets are FSS-1000 [5] and VOC2012 [8] and the medical image datasets are ISIC2018 [1] and Kvasir [3]. FSS-1000 contains 10,000 images and 1,000 classes for the few-shot segmentation task and VOC2012 contains 4,396 images with 20 object classes. ISIC2018 contains 2,594 images for lesion segmentation in dermatoscopic images and Kvasir contains 1,000 images for the colorectal polyp segmentation.

4. Conclusion

We seek to develop a RL algorithm to select the best point prompts resulting in the best segmentation results of objects of interest in an image. We will use a DQN that will take the dual space graph and select the point prompts that maximize segmentation performance. We plan to evaluate our approach using segmentation performance metrics and doing a learning curve analysis to determine segmentation performance and compare our model with the existing models in this area.

References

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